SEC P vs NP log 2026-05-17

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-17 15:23 UTC

SEC P vs NP — daily log 2026-05-17

25 cycles recorded

GCT Lie-Stabilizer Codimension Linearly Bounds ACC^0 Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Geometric Complexity Theory: Lie-algebra stabilizers of multilinear polynomial extensions in the Mulmuley-Sohoni-Bürgisser-Ikenmeyer framework × $ACC^0[m]$ circuit size for Boolean functions (Sipser-like targets)
- **Recorded**: 2026-05-17 00:01 UTC
- **Entry ID**: `83d26c2ef069`

Statement

For a Boolean function f:{-1,+1}^n -> {-1,+1}, let F_f(x) = Σ_{S⊆[n]} f̂(S) · Π_{i∈S} x_i be its multilinear Walsh extension over Q, let L(F_f) := {A ∈ gl_n(Q) : Σ_{i,j} A_{ij} x_j ∂_i F_f ≡ 0 in Q[x_1,…,x_n]} be its GCT Lie stabilizer, and set γ(f) := n^2 − dim_Q L(F_f). Conjecture: there is an absolute constant C ≤ 10 such that every $ACC^0[m]$ circuit of size s computing f satisfies γ(f) ≥ n^2 − C·s. A single $ACC^0[m]$ circuit of size s computing some f with γ(f) < n^2 − 10·s falsifies the conjecture.

Rationale

GCT's symmetry-characterization principle says polynomials whose stabilizer Lie algebras exceed generic dimension are atypical; we invert this to claim small ACC^0 circuits cannot accidentally synthesize multilinear extensions of anomalously high symmetry. The Lie stabilizer is GL_n-equivariant rather than truth-table-dense, so it dodges Razborov–Rudich's 'large + constructive' clause, and it is computed from the GL_n action on polynomials, not from black-box algebraic extensions — placing the invariant outside the algebrization and relativization regimes that GCT was designed to escape.

Novelty

- Judge: `NOVEL` over 11 arXiv hits

Top hits consulted: - [1212.5380v2] On properties of principal elements of Frobenius Lie algebras - [1606.05050v1] Proof Complexity Lower Bounds from Algebraic Circuit Complexity - [2411.11095v3] Invariant theory and coefficient algebras of Lie algebras - [1904.05483v2] Parallels Between Phase Transitions and Circuit Complexity? - [1504.04131v2] Formulas for the Walsh coefficients of smooth functions and their application to bounds on the Walsh coefficients

Empirical Test

- exit code: `1`, elapsed: `0.11s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_698adb63.py", line 102, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_698adb63.py", line 86, in run_trial
    gamma = gaussian_elimination(system)
            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_698adb63.py", line 65, in gaussian_elim
    if matrix[i][i] == 0:
       ~~~~~~~~~^^^
IndexError: list index out of range
```

Judge reasoning

The test crashed with an IndexError in gaussian_elimination before any of the 480 trials or reference-function checks produced slack data, so the pre-registered support condition cannot be evaluated. Per Rule 2, a crash yields INCONCLUSIVE. | next: Fix the Gaussian elimination routine (guard against out-of-range pivot indexing and handle rectangular/short matrices) and rerun the full (n,s)-grid plus reference-function suite to obtain actual slack statistics.

2-adic Newton Slopes of DISJ Sub-blocks Scale as √n

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: p-adic analysis (Newton polygons over Q_2 of integer characteristic polynomials) × Randomized communication complexity of DISJOINTNESS via average-case tensor invariants on sub-blocks
- **Recorded**: 2026-05-17 00:26 UTC
- **Entry ID**: `efd166108501`

Statement

Let M_n ∈ {-1,+1}$^{\{2}$n × 2^n} be the signed disjointness matrix M_n[x,y] = (-1)^{[x∩y≠∅]} where x,y are interpreted as subsets of [n]. For each n, draw uniformly random row set R and column set C with |R|=|C|=k=min(2n,40), form A := M_n[R,C], compute its exact integer characteristic polynomial p(t)=Σ_{i=0}^{k} c_i t^i, and define the 2-adic Newton slope count ν_2(A) := number of distinct slopes in the lower convex hull of {(i, v_2(c_i)) : c_i ≠ 0} where v_2 is the 2-adic valuation (v_2(0):=+∞). Conjecture: E_{R,C}[ν_2(M_n^{R,C})] = Θ(√n); moreover for every n ≥ 8 the DISJ median exceeds the median of ν_2 for an i.i.d. ±1 matrix of the same shape by a factor ≥ 1.5, and a single (n, seed) pair where this median ratio inverts falsifies the conjecture.

Rationale

The subset semilattice underlying DISJ injects 2-adic congruence structure into every sub-block (parity of intersection size acts like a Z_2-valued character), so subdeterminants accumulate distinguished 2-adic valuations that random Bernoulli matrices lack — a property that should propagate, via Yao's principle, into a lower bound on randomized communication. Newton polygons over Q_2 are not preserved by polynomial extension to F_2[t]/⟨t^2-t⟩, evading the algebrization barrier, and the invariant is structural rather than 'large+useful on truth tables', evading Razborov-Rudich. The bridge between p-adic Newton-polygon geometry and communication complexity has essentially no prior literature, while the average-to-worst-case framing (E[ν_2] ↦ worst-case CC lower bound) follows the Impagliazzo-Wigderson template.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 34.76s

```
tances_tested': 5, 'conjecture_holds': False, 'counterexample': 'Ratio 1.0 < 1.5 for seed 547'}
TRIAL: {'metric_name': 'median_newton_slope_count', 'metric_value': 2, 'instances_tested': 5, 'conje
TRIAL: {'metric_name': 'median_newton_slope_count', 'metric_value': 2, 'instances_tested': 5, 'conje
TRIAL: {'metric_name': 'median_newton_slope_count', 'metric_value': 2, 'instances_tested': 5, 'conje
TRIAL: {'metric_name': 'median_newton_slope_count', 'metric_value': 2, 'instances_tested': 5, 'conje
TRIAL: {'metric_name': 'median_newton_slope_count', 'metric_value': 2, 'instances_tested': 5, 'conje
TRIAL: {'metric_name': 'median_newton_slope_count', 'metric_value': 2, 'instances_tested': 5, 'conje
TRIAL: {'metric_name': 'median_newton_slope_count', 'metric_value': 2, 'instances_tested': 5, 'conje
TRIAL: {'metric_name': 'median_newton_slope_count', 'metric_value': 2, 'instances_tested': 5, 'conje

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b68324ac.py", line 136, in <module>
    failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
                   ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b68324ac.py", line 136, in <genexpr>
    failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
                        ~^^^^^^^^
KeyError: 'seed'
```

Judge reasoning

Safety rail: critic_challenge_falsified | original: Multiple trials report conjecture_holds=False with ratio 1.0 < 1.5 (e.g., seeds 547, 593, 631, ...), satisfying the pre-registered falsification condi | next: Re-run with corrected seed-logging, stratify by $n \in \{10,14,18,22\}$, and check whether the ν_2 statistic saturates at small k=min(2n,40) — perhaps a normalized slope-density (ν_2/k) would better capture any DISJ-vs-random gap.

Cyclic Burnside Orbit Count Lower-Bounds AC^0 Size for PARITY

- **Verdict**: INCONCLUSIVE
- **Bridge**: Burnside-ring / equivariant orbit counting of cyclic group actions on labeled circuit DAGs (a corner of combinatorial representation theory rarely applied to circuit complexity) × AC^0 circuit size for PARITY_n, with the lifting-style intuition that gadget symmetry constrains the lifted communication matrix
- **Recorded**: 2026-05-17 00:49 UTC
- **Entry ID**: 80d12cf7a27e

Statement

Let C be an AC^0 circuit over inputs x_0,...,x_{n-1} computing PARITY_n with n a prime ≥ 3, depth(C)=d and size(C)=s, and let σ be the cyclic shift σ(i)=i+1 mod n acting on each gate g of C by simultaneously relabeling every input wire x_i appearing in g's recursive expansion to x_{σ(i)}. Define ψ(C) as the number of ⟨σ⟩-orbits of gates of C under the equivalence g ∼ h iff some σ^k maps the lex-min canonical (op-type, sorted-children) form of g to that of h. Then for every such C computing PARITY_n we conjecture ψ(C) ≥ (1/(4d)) · log_2(s), refuted by any single PARITY-computing AC^0 circuit whose ratio 4·d·ψ(C)/log_2(s) drops below 1.

Rationale

Lifting theorems translate query lower bounds into communication lower bounds by composing the function with a symmetry-breaking gadget; dually, when the target function (PARITY) is itself shift-invariant, a small circuit must reuse few genuinely distinct sub-computations, so the Z_n-orbit count of its gates becomes a Burnside-style proxy for the work that random restrictions are usually invoked to expose. Because ψ is defined on the circuit DAG rather than on the truth table, the invariant sits outside the Razborov–Rudich natural-proofs barrier; because the construction is purely combinatorial (no polynomial extension of the Boolean ring) it does not algebrize, matching the LIFTING approach's safe-direction profile.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [2304.02770v2] Tight Correlation Bounds for Circuits Between AC0 and TC0 - [2504.19966v3] Quantum circuit lower bounds in the magic hierarchy - [1708.00951v2] Canonical Heights and Monomial Maps: On Effective Lower Bounds for Points with Dense Orbit - [1904.05483v2] Parallels Between Phase Transitions and Circuit Complexity? - [2005.06373v2] Counting Schur Rings over Cyclic Groups of Semi-prime Order

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c17bcfb6.py", line 225, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c17bcfb6.py", line 180, in run_trial
    psi = compute_psi(circuit, n)
          ^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c17bcfb6.py", line 139, in compute_psi
    gate_hashes[gate] = hash_gate(gate)
                        ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c17bcfb6.py", line 126, in hash_gate
    return hash(gate)
           ^^^^^^^^^^
TypeError: unhashable type: 'list'
```

Judge reasoning

The test crashed with a TypeError (unhashable list) in hash_gate before any trial completed, so no r values were produced and the pre-registered support condition could not be evaluated. | next: Fix the canonicalization in hash_gate by converting children lists to tuples (or using a frozen canonical form) before hashing, then re-run the full 2160-circuit grid.

Cauchy Mean Width of Minterm Hull Lower-Bounds Monotone k-CLIQUE

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Convex geometry (Cauchy mean-width / intrinsic 1-volume of polytopes in R^N, in the spirit of Schneider's Brunn–Minkowski theory, rarely connected to circuit complexity) × Karchmer–Wigderson / monotone DNF size and formula complexity for the k-CLIQUE indicator
- **Recorded**: 2026-05-17 01:28 UTC
- **Entry ID**: `65d1ffdf3cc6`

Statement

For a monotone Boolean function f on N variables, let M(f) ⊂ {0,1}^N be its set of minterms (minimal satisfying assignments viewed as 0/1 vectors) and let K(f) := conv(M(f) ∪ {0}) ⊆ [0,1]^N. Define μ(f) := MW(K(f))^2, where MW(K) = 2·E_{u~Unif(S^{N−1})}[max_{x∈K}⟨u,x⟩] is the Cauchy mean width. We conjecture: (i) for every monotone DNF representation of f with s terms, μ(f) ≤ C_1·log(s+1)·(log N+1); (ii) for the k-CLIQUE indicator on K_v with k=⌈$\log_2$ v⌉ (so N = v(v−1)/2), μ(f) ≥ C_2·v, for absolute constants C_1,C_2>0. A single instance with μ(f) > C_1·log(s+1)(log N+1), or a k-CLIQUE indicator with μ < C_2·v at v ≤ 9, falsifies the conjecture.

Rationale

Monotone DNFs are unions of axis-aligned subcubes, so conv(M(f)) is the convex hull of at most s lattice vectors of bounded coordinate sum, and its mean width should grow only logarithmically with the number of generators (a Sudakov/Talagrand-style upper bound for sparse 0/1 hulls). The k-CLIQUE minterm cloud, in contrast, lives on the (k choose 2)-th slice and is rotationally rich because v-vertex automorphisms act transitively on edges, forcing the supporting hyperplanes to spread evenly over many directions and inflating mean width linearly in v. If both bounds hold, any monotone DNF for k-CLIQUE has s ≥ 2^{Ω(v/log v)}, a Razborov-flavor separation via a purely metric-geometric witness that bypasses approximator combinatorics.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `0`, elapsed: `33.94s`


```
me": "mean_width_squared", "metric_value": 1.136003662687513, "instances_tested": 17, "conjecture_ho
CLIQUE with v=4 has mu=1.136003662687513 < 0.5*v=2.0", "seed": 421}
TRIAL: {"metric_name": "mean_width_squared", "metric_value": 1.1857866298955364, "instances_tested"
CLIQUE with v=4 has mu=1.1857866298955364 < 0.5*v=2.0", "seed": 463}
TRIAL: {"metric_name": "mean_width_squared", "metric_value": 1.1927318541604457, "instances_tested"
CLIQUE with v=4 has mu=1.1927318541604457 < 0.5*v=2.0", "seed": 503}
TRIAL: {"metric_name": "mean_width_squared", "metric_value": 1.1396899464107255, "instances_tested"
CLIQUE with v=4 has mu=1.1396899464107255 < 0.5*v=2.0", "seed": 547}
TRIAL: {"metric_name": "mean_width_squared", "metric_value": 1.1832069622303316, "instances_tested"
CLIQUE with v=4 has mu=1.1832069622303316 < 0.5*v=2.0", "seed": 593}
TRIAL: {"metric_name": "mean_width_squared", "metric_value": 1.1742179582193337, "instances_tested"
CLIQUE with v=4 has mu=1.1742179582193337 < 0.5*v=2.0", "seed": 631}
TRIAL: {"metric_name": "mean_width_squared", "metric_value": 1.1635078759135067, "instances_tested"
CLIQUE with v=4 has mu=1.1635078759135067 < 0.5*v=2.0", "seed": 677}
TRIAL: {"metric_name": "mean_width_squared", "metric_value": 1.1855075477194135, "instances_tested"
CLIQUE with v=4 has mu=1.1855075477194135 < 0.5*v=2.0", "seed": 727}
TRIAL: {"metric_name": "mean_width_squared", "metric_value": 1.172332125064841, "instances_tested":
CLIQUE with v=4 has mu=1.172332125064841 < 0.5*v=2.0", "seed": 773}
TRIAL: {"metric_name": "
```


Judge reasoning

Safety rail: critic_challenge_falsified | original: Every k-CLIQUE trial at v=4 yields μ ≈ 1.17, well below the falsification threshold 0.25·v = 1.0… actually 0.25·4=1.0 so this passes the 0.25·v floo | next: Rescale the lower bound to μ(f) ≥ C_2·v/log v or test the conjecture with normalized mean width MW(K(f))/√N to remove metric-saturation artifacts at small v.

Costas Displacement Coincidence Lower-Bounds AC^0 Size for PARITY

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Costas array / Welch–Costas displacement combinatorics (Golomb–Costas radar arrays and B_2/Sidon-deficiency theory of integer sets) — a corner of additive combinatorics developed for radar/sonar that has essentially no presence in circuit complexity literature × AC^0 depth-d circuit size for PARITY_n (Hastad-style lower bounds, viewed through the lens of bottom-gate support designs)
- **Recorded**: 2026-05-17 02:01 UTC
- **Entry ID**: `6bb89da9ec2e`

Statement

Let C be a depth-d AC^0 circuit on n inputs with bottom-level (depth-1) AND/OR gates having literal-supports S_1,…,S_m ⊆ [n]. Define the Costas displacement defect κ(C) = log_2(1 + max_{δ ∈ {1,…,n-1}} Σ_{i=1}^{m} |{(a,b) ∈ S_i × S_i : a > b and a−b ≡ δ (mod n)}|). We conjecture that for every C that computes PARITY_n exactly, κ(C) ≥ (1/4) · n^{1/(d−1)}, and consequently size(C) ≥ 2^{Ω(κ(C))}. A single depth-d AC^0 circuit that computes PARITY_n while having κ(C) < (1/4) · n^{1/(d−1)} refutes the conjecture.

Rationale

PARITY is translation-invariant on the Boolean cube, so any AC^0 circuit computing it must spread its bottom-level supports densely across all cyclic displacements — exactly the property that Costas/Welch arrays minimize. Replacing Hastad's random restriction by a Costas-style design-theoretic lower bound turns AC^0 PARITY bounds into a question about Sidon-deficiency of support multisets. The invariant depends only on the circuit's bottom-support combinatorics (not the function's truth table), so it sidesteps Natural Proofs, and uses no ring operations, sidestepping algebrization.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `240.18s`

```
TIMEOUT after 240s
```

Judge reasoning

The test timed out after 240s (returncode 124) without producing a RESULT line, so neither the support nor falsification condition could be evaluated. | next: Reduce the parameter sweep (e.g., cap n at 20 and d=2) or parallelize per-seed evaluation to produce data within the timeout.

Vertex Star-Discrepancy Equals Spectral Norm for XOR Comm

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Geometric discrepancy theory (vertex-restricted L∞ star discrepancy of {0,1}^n point sets, in the Chazelle–Matousek tradition of axis-parallel box discrepancy) × Randomized two-party communication complexity of XOR-functions f(x,y)=g(x⊕y) accessed through the discrepancy-method lower bound disc(M_f)=||ĝ||_∞
- **Recorded**: 2026-05-17 02:29 UTC
- **Entry ID**: `d27e7e45b29b`

Statement

For any non-constant g:{0,1}^n→{−1,+1}, let P_g={z∈{0,1}^n : g(z)=−1} and define the lattice star discrepancy D*lat(g) := max{a∈{0,1}^n} | |P_g ∩ [0,a]⊓|/2^n − (|P_g|/2^n)·(2{popcount(a)}/2^n) |, where [0,a]⊓={z∈{0,1}^n : z≤a coordinatewise}; let ||ĝ||∞ := max{S⊆[n]} |ĝ(S)| be the spectral norm. Conjecture: there exist absolute constants c_1=1/8 and c_2=8 such that for every n≥4 and every non-constant g, c_1·||ĝ||_∞ ≤ D*lat(g) ≤ c_2·||ĝ||∞·log_2(n+1). A single (n,g) pair violating either inequality refutes the conjecture.

Rationale

The discrepancy lower bound on randomized XOR-communication is exactly log(1/||ĝ||∞), yet ||ĝ||∞ is a global Fourier quantity with no geometric face. Vertex star discrepancy is the canonical Chazelle–Matousek irregularity measure of a point set inside [0,1]^n; restricting test boxes to lattice corners makes it computable in O(n·2^n) via the zeta transform. If D*lat ≈ ||ĝ||∞, this gives the first concrete geometric (axis-parallel box) certificate of XOR-communication hardness, opening a Banaszczyk/Kashin-style geometric attack on unsaturated R(DISJ)-like bounds.

Novelty

- Judge: `NOVEL` over 2 arXiv hits

Top hits consulted: - [s2:1704.02537] Dual polynomials and communication complexity of XOR functions - [s2:19311e592db3bd54ad8685ae0d27b815c87be] Dual polynomials and communication complexity of XOR functions

Empirical Test

- exit code: `124`, elapsed: `240.03s`

```
TIMEOUT after 240s
```

Judge reasoning

The test timed out after 240s (returncode=124) without producing a RESULT line or any trial data, so neither the support nor falsification condition can be evaluated. | next: Reduce the maximum n or restrict the enumeration over a∈{0,1}^n (e.g., sample or use a smarter combinatorial bound) so the 210 trials complete within the time budget.

Operator-SoS 4-Trace Gap Lower-Bounds Blocks-Order Read-Twice BP for IP_2

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Sum-of-Squares / Lasserre hierarchy on operator-valued layer-product tensor moments (Putinar–Helton-style non-commutative SoS pseudo-trace invariants applied to layer-difference matrices of branching programs — a corner combining moment-SoS literature with BP structural invariants, with <5 papers on the BP side) × Read-twice oblivious branching program size for IP_2 (the open BP_READTWICE separation of Borodin–Razborov–Smolensky 1993)
- **Recorded**: 2026-05-17 02:59 UTC
- **Entry ID**: `0ecd5f06ea81`

Statement

Let P be a read-twice oblivious BP on 2n Boolean variables (x_1,…,x_n,y_1,…,y_n) with width w and length 4n, layer transition matrices M_j(0),M_j(1)∈{0,1}^{w×w}, and layer-difference matrices

$N_j := M_j(1)-M_j(0)$; for each variable v with its two read-layers (a_v,b_v) form the paired operator $P_v := N_{a_v}N_{b_v}\in\mathbb{R}^{w\times w}$, and define the operator-SoS 4-trace gap $\rho(P) := \log_2 \max_{u\neq v} |\langle P_u, P_v\rangle_F| / w^2$, where $\langle A,B\rangle_F = \mathrm{tr}(A B^T)$. CONJECTURE: (Upper anchor, trivial via SoS-Cauchy-Schwarz) $\rho(P) \le 2\cdot\log_2 \mathrm{width}(P) + 2$ for every read-twice oblivious BP P; (Lower bound, non-trivial) for every BLOCKS-ORDERED read-twice oblivious BP P (all x-layers preceding all y-layers in each of the two reads) that computes $IP_2(x,y)=\oplus_i x_i y_i$ with bias $\ge 1/2 + n^{-2}$, $\rho(P) \ge n/16 - 4\cdot\log_2 n$. Equivalently, any such blocks-ordered read-twice BP for IP_2 has width $\ge 2^{n/40}$ for all sufficiently large n; a single small-width blocks-ordered read-twice BP for IP_2 with $\rho(P) < n/16 - 4\cdot\log_2 n$ falsifies the conjecture.

Rationale

The paired operator $P_v = N_{a_v} N_{b_v}$ packages the two reads of variable v into a single non-commutative SoS-style moment, and the Frobenius pairing $\langle P_u,P_v\rangle_F$ is exactly the kind of trace identity that survives Lasserre relaxation at level 2 yet remains computable directly from the BP. For IP_2 the algebraic obstruction that forces exponential read-once size — namely that x and y blocks cannot be summarized in fewer than 2^n states without bilinear cancellation — should reappear in the read-twice setting as a forced spread of mass across many variable-pairs in the Gram matrix of $\{P_v\}$, producing a heavy off-diagonal entry. Because ρ depends on the specific BP layer matrices (not the truth table) it cannot collapse to a function-symmetric invariant, side-stepping the natural-proofs trap that killed the prior Lasserre-rank attempt in this approach.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8f1e5764.py", line 113, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8f1e5764.py", line 69, in run_trial
    N = generate_bp(w, n, seed)
        ^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8f1e5764.py", line 44, in generate_bp
    N = [matrix_subtract(M[j], M[j-1]) if j > 0 else M[j] for j in range(2*n)]
         ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8f1e5764.py", line 28, in matrix_subtra
    w = len(A)
        ^^^^^^
TypeError: object of type 'int' has no len()
```

Judge reasoning

The test crashed with a TypeError in matrix_subtract before producing any data, so neither the upper-anchor nor the IP_2 lower-bound criterion was evaluated. No RESULT line was emitted and the pre-registered support condition cannot be assessed. | next: Fix the matrix representation bug in generate_bp/matrix_subtract (ensure M[j] is a 2D list/array of shape $w\times w$, not an int) and re-run the 540 random instances plus the 4 explicit IP_2 BPs.

Cross-Side Fourier Symmetric-Difference Mass Lower-Bounds DISJ CC_R

- **Verdict**: `BARRIER_HIT`
- **Bridge**: Fourier analysis of Boolean functions (Bonami–Beckner / KKL-style level-weighting on the cube, with bipartite cross-coordinate noise operators rarely used in CC) × Randomized two-party communication complexity of DISJOINTNESS (CC_R) and other partial functions on N=2^n × 2^n matrices
- **Recorded**: 2026-05-17 03:06 UTC
- **Entry ID**: `fba37ecf9a43`

Statement

For a Boolean matrix M ∈ {0,1}$^{\{2}$n × 2^n} viewed as a function f_M : {0,1}^{2n} → {0,1} with first n bits the x-input and last n bits the y-input, and for T ⊆ [2n] let T_x = T ∩ [n] and T_y = {i ∈ [n] : i+n ∈ T}, define δ(T) := |T_x △ T_y| and the cross-asymmetry invariant Ψ(f_M) := (Σ_T f_M_hat(T)^2 · 2^{δ(T)}) / (Σ_T f_M_hat(T)^2). Conjecture: there exists an absolute constant c ≥ 1/8 such that for every Boolean matrix M, CC_R^{1/3}(M) ≥ c · log_2 Ψ(f_M), and moreover Ψ(f_{DISJ_n}) = (7/6)^n exactly (so log_2 Ψ = n·log_2(7/6) ≈ 0.222 n and CC_R(DISJ_n) = Ω(n) follows). A single Boolean matrix M with a verified protocol of cost B and log_2 Ψ(f_M) > 8·B refutes the conjecture.

Rationale

Standard Fourier □_1 / spectral norm lower bounds for CC are oblivious to the bipartition structure of the input, while CC_R cares essentially about which Fourier modes 'mix' the two parties. Weighting each Fourier mass by 2^{|T_x △ T_y|} amplifies modes that genuinely couple x- and y-coordinates asymmetrically — EQ-type matrices live on the diagonal T_x = T_y and get Ψ = 1, while DISJ's product-over-coordinates structure forces a multiplicative gain of 7/6 per coordinate, giving the right Razborov-type Ω(n). The hypothesis is therefore a Fourier-analytic 'cross-side' refinement of the standard discrepancy/spectral-norm method, in the spirit of KKL-style coordinate-weighted Fourier inequalities.

Novelty

- Judge: " over 0 arXiv hits

Empirical Test

- exit code: `-1`, elapsed: `0.00s`

```
(no output)
```

Judge reasoning

Rejected by NATURAL_PROOFS barrier

---

q-Major Cancellation Gap Separates Permanent from Determinant Supports

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Mahonian statistics on the symmetric group (major index, q-factorials, RSK cocharge) × Padded permanent/determinant comparison on shared 0/1 support matrices (GCT det-vs-perm flavor)
- **Recorded**: 2026-05-17 03:32 UTC
- **Entry ID**: `796531c3b7fa`

Statement

For an n x n matrix M in {0,1}^{n x n}, let T_M = {sigma in S_n : M[i,sigma(i)] = 1 for all i} and define the q-major permanent and signed q-major determinant on the support T_M by perm_q(M) := sum_{sigma in T_M} q^{maj(sigma)} and det_q(M) := sum_{sigma in T_M} sign(sigma) * q^{maj(sigma)}, where maj(sigma) is the major index (sum of descent positions). Conjecture: for every n >= 3 and every 0/1 matrix M with |T_M| >= 2, the q=2 cancellation ratio R_2(M) := perm_2(M) / max(1, |det_2(M)|) satisfies R_2(M) >= sqrt(|T_M|) / (2*n). Equivalently, the sign(sigma)-weighted Mahonian sum on any 0/1-support exhibits at least square-root-of-support cancellation against the unsigned version, up to a linear factor in n.

Rationale

Permanent vs determinant in GCT differs precisely by replacing sign(sigma) with 1 inside the S_n-sum; weighting both sums by q^{maj(sigma)} (a Schur-positive Hecke-algebra principal-specialization) refracts that difference through irreducible S_n-character spectra, and at q=2 one already sees [n]_2! = prod(2^k - 1) for perm_n but a much smaller signed sum for det_n. A square-root cancellation lower bound on every common support T_M would certify that determinantal coefficient sequences on permanental supports always look 'random-like' relative to their unsigned partners, giving a Mahonian witness for the per-vs-det gap that is preserved under support-preserving substitutions.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [1503.00822v2] Product Ranks of the 3×3 Determinant and Permanent - [2302.11221v6] A q-analog of certain symmetric functions and one of its specializations - [1302.4419v3] The determinant of the Malliavin matrix and the determinant of the covariance matrix for multiple integrals - [0810.2731v3] Fix-Euler-Mahonian statistics on wreath products - [1101.4332v2] Mahonian Pairs

Empirical Test

- exit code: `1`, elapsed: `1.37s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c5e11f1c.py", line 51, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c5e11f1c.py", line 33, in run_trial
    perm_q = sum(2 ** sum(i for i, x in enumerate(sigma) if sigma[i] > sigma[j]) for sigma in permutation
             ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c5e11f1c.py", line 33, in <genexpr>
    perm_q = sum(2 ** sum(i for i, x in enumerate(sigma) if sigma[i] > sigma[j]) for sigma in permutation
                          ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c5e11f1c.py", line 33, in <genexpr>
    perm_q = sum(2 ** sum(i for i, x in enumerate(sigma) if sigma[i] > sigma[j]) for sigma in permutation
                                                                                  ^
NameError: name 'j' is not defined
```

Judge reasoning

The test crashed with a NameError ('j' undefined) in the major-index computation before any trial completed, so no data was produced to evaluate the pre-registered criterion. | next: Fix the maj(sigma) computation to correctly count descent positions (i where sigma[i] > sigma[i+1]) and rerun the 240-trial sweep.

Immanant Positivity Width Bounds Monotone KW Depth for Perfect Matching

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Stembridge–Stanley immanant theory: the λ-immanants imm_λ(S) = Σ_σ χ_λ(σ) Π_i S[i,σ(i)] organise matchings of a 0/1 matrix into Schur–Weyl S_n-isotypic cells (a corner of symmetric-group representation theory with very few papers connecting it to monotone circuit lower bounds outside of GCT speculation) × Monotone Karchmer–Wigderson tree depth d_KW(S) at a 1-input S of the bipartite perfect-matching function PM_n, equivalent by KW to log_2 of the local monotone formula leaf size restricted to matchings inside S vs. small bipartite vertex covers
- **Recorded**: 2026-05-17 04:06 UTC
- **Entry ID**: `15b45d9e4966`

Statement

For 3 ≤ n ≤ 6 and any n×n 0/1 matrix S with perm_n(S) ≥ 1, let Σ(S) := {σ ∈ S_n : S[i,σ(i)] = 1 ∀i} and define the immanant positivity width w(S) := #{λ ⊢ n : imm_λ(S) > 0}. Let d_KW(S) be the minimum depth of any KW protocol that on Alice-input τ ∈ Σ(S) and Bob-input C, an (n−1)-vertex cover of the bipartite complement of some 0-input S′, outputs an edge (i, j) ∈ τ C. Conjecture: d_KW(S) ≥ ⌈log_2 w(S)⌉ for every such S, with equality only when Σ(S) is RSK-shape-monochromatic; in particular w(I_n) = p(n), forcing d_KW(I_n) ≥ ⌈log_2 p(n)⌉ = Θ(√n), a representation-theoretic obstruction in the spirit of GCT's perm/det separation.

Rationale

Each rectangle of a KW protocol pools matchings that all hit a common edge of Bob's cover; we conjecture this pooling is incompatible with carrying more than one irreducible χ_λ-component, so the number of non-vanishing λ-immanants is a Schur-Weyl obstruction to small KW trees. This mirrors the GCT philosophy that perm_n's orbit closure is Schur-rich while det_m's is sign-isotypic: a small substitution-stable Schur-width invariant would translate directly into a perm-vs-det monotone gap. The link of immanants to KW depth is, to our knowledge, untouched in the literature.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [2107.05128v2] Karchmer-Wigderson Games for Hazard-free Computation - [2002.07444v1] Multiparty Karchmer-Wigderson Games and Threshold Circuits - [2003.09602v2] The Number of Perfect Matchings in Möbius Ladders and Prisms - [2304.05285v1] Hook-Shape Immanant Characters from Stanley-Stembridge Characters - [1801.08709v2] Adaptive Lower Bound for Testing Monotonicity on the Line

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_70051383.py", line 178, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_70051383.py", line 132, in run_trial
    d_KW = kw_depth(matrix, covers)
           ^^^^^^^^^^^^^^^^^^^^^^^^
```

```
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_70051383.py", line 104, in kw_depth
    subsets[cover].append(sigma)
    ~~~~~~~^^^^^^^
TypeError: unhashable type: 'set'
```

Judge reasoning

The test crashed with a TypeError ('unhashable type: set') in kw_depth before producing any data, so no metrics were computed and the pre-registered support condition cannot be evaluated. | next: Fix the kw_depth implementation by using a hashable key (e.g., frozenset) for cover keys in the subsets dict, then rerun the 360-matrix sweep.

Curto-Fialkow Hankel Defect Separates Read-Twice IP_2 BPs

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Curto-Fialkow truncated moment problem theory (Hankel positivity and flat-extension rank, the real-algebraic-geometry foundation of the Lasserre SoS hierarchy; Curto-Fialkow 1996, Lasserre 2001) applied to layer-conditional empirical moments of branching programs × Read-twice oblivious branching program size for IP_2 on 2n variables in adversarial order x_1,…,x_n,y_1,…,y_n (Borodin-Razborov-Smolensky 1993 open separation)
- **Recorded**: 2026-05-17 04:32 UTC
- **Entry ID**: `9f09553bfd37`

Statement

Let P be a read-twice oblivious BP with state set Q (|Q|=s) computing a Boolean function on N=2n variables, and let q(z,□) denote its state on input z after □ steps. For each state q∈Q and each layer □∈{1,…,2N}, define the centered conditional moment sequence m^{q,□}*k(P) = 2^{-N} Σ_z □[q(z,□)=q]·(IP_2(z)-1/2)^k for k=0,…,⌊N/4⌋, and let H^{q,□}(P) be the (d+1)×(d+1) Hankel matrix with d=⌊N/8⌋ and entries m^{q,□}*{i+j}. Define the positivity defect δ(P) = max_{q,□} #{negative eigenvalues of H^{q,□}(P)}. Conjecture: (i) for every read-twice oblivious BP P of size s, δ(P) ≤ 8·⌈log_2(s+1)⌉; (ii) every read-twice oblivious BP P that exactly computes IP_2 in the adversarial schedule satisfies δ(P) ≥ ⌈√N/2⌉.

Rationale

Curto-Fialkow theory equates degree-2d SoS feasibility with the existence of flat positive extensions of truncated Hankel sequences, so a small BP forces low-rank flat extensions of its layer-conditional empirical moments — a structural constraint never previously exploited for BP lower bounds. IP_2's bilinear ⊕-structure should force every conditional sub-population reaching a given state at a given layer to retain non-trivial bilinear correlation between the unread x-bits and the moments, obstructing flat positive Hankel extension and producing many negative eigenvalues. The conjectured gap log(s) vs √N would yield s ≥ 2^{Ω(√N)} for read-twice IP_2, attacking the BRS-1993 problem via the SoS-moment side rather than via communication or partition arguments.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
```

```
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8925cd5b.py", line 155, in <module>
    result = run_trial(int(seed))
             ^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8925cd5b.py", line 104, in run_trial
    P, Q, N = generate_random_bp(n, w, seed)
              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8925cd5b.py", line 64, in generate_rand
    P[(q, l)][k] += 1
    ~~~~~~~~~^^^
TypeError: list indices must be integers or slices, not float
```

Judge reasoning

The test crashed with a TypeError in generate_random_bp before producing any data, so neither the support nor falsification condition could be evaluated. | next: Fix the indexing bug in generate_random_bp (cast the index k to int) and rerun the pre-registered sweep.

Hodge-Cheeger Product Lower-Bounds Tseitin Resolution Length

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Combinatorial Hodge theory on graphs (pairing of harmonic 1-cochain dimension b_1 with vertex-Laplacian algebraic connectivity, in the Lim 2020 / Jiang-Lim-Yao-Ye framework used in TDA but essentially unused in proof complexity) × Tree-like Resolution refutation length of Tseitin XOR formulas T(G,omega) on bounded-degree graphs G with odd charge omega
- **Recorded**: 2026-05-17 05:07 UTC
- **Entry ID**: `b60e3f3f26e1`

Statement

Let G=(V,E) be a connected graph with max degree d_max <= 6 and |V|=n, and let omega:V->F_2 satisfy sum_v omega_v = 1 (mod 2). Define b_1(G) := |E|-|V|+1 (the first combinatorial Betti number, i.e. cycle-space dimension over F_2), let lambda_2(L_G) denote the second-smallest eigenvalue of the combinatorial vertex Laplacian L_G=D-A, and set the Hodge-Cheeger invariant nu(G) := lambda_2(L_G) * b_1(G) / d_max. Conjecture: there is an absolute constant c >= 0.5 such that every tree-like Resolution refutation of T(G,omega) has length at least 2^{c*nu(G)}; a single (G,omega) with nu(G) >= 5 and a Resolution refutation of length < 2^{0.5*nu(G)} falsifies the conjecture.*

Rationale

Tseitin hardness needs two ingredients: enough independent cycles to encode unsatisfiable XOR obstructions (measured by b_1) and enough spectral mixing to prevent localized cancellation (measured by lambda_2); combinatorial Hodge theory packages exactly this pairing via the orthogonal decomposition of edge flows into gradients and harmonic 1-cochains. Bridges/paths zero out b_1; bottlenecked graphs zero out lambda_2; only graphs that are simultaneously cycle-rich and spectrally expanding (the BSW regime) yield large nu, matching the empirical hard-Tseitin profile. The framing is novel for proof complexity (<5 papers connect Hodge theory to Resolution) yet entirely poly-time computable via a single numpy eigendecomposition.

Novelty

- Judge: `NOVEL` over 10 arXiv hits

Top hits consulted: - [2209.05839v3] On bounded depth proofs for Tseitin formulas on the grid; revisited - [2103.09609v1] Characterizing Tseitin-formulas with short regular resolution refutations

- [0906.0693v3] An improved lower bound on the counterfeit coins problem - [1301.2045v3] Integral-valued polynomials over the set of algebraic integers of bounded degree - [1606.05050v1] Proof Complexity Lower Bounds from Algebraic Circuit Complexity

Empirical Test

- exit code: `124`, elapsed: `240.10s`

```
TIMEOUT after 240s
```

Judge reasoning

The test timed out after 240s (returncode 124) before producing any RESULT line or data, so the pre-registered support condition cannot be evaluated. | next: Reduce instance count or graph sizes (e.g., shrink to 3 categories × 3 sizes × 10 seeds) and/or impose a per-instance Resolution-search timeout so the suite completes within the wall-clock budget.

Sandpile Group Order Lower-Bounds Monotone KW Depth

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Abelian sandpile / chip-firing group theory (Dhar 1990, Bjorner-Lovasz-Shor 1991; the critical/sandpile group K(G) = coker of the reduced graph Laplacian, |K(G)| = number of spanning trees by Kirchhoff). This corner of algebraic combinatorics / statistical mechanics has essentially no presence in monotone circuit lower-bound literature. × Monotone Karchmer-Wigderson protocol depth D_KW(f) = log_2 of minimum monotone formula leaf size for a monotone Boolean function f, accessed via the bipartite KW relation (minterm, maxterm) -> intersection coordinate.
- **Recorded**: 2026-05-17 05:45 UTC
- **Entry ID**: `f75d64a69735`

Statement

For every nonconstant monotone Boolean f:{0,1}^n -> {0,1} let M(f) and K(f) be its minterm and maxterm antichains (as subsets of [n]). Form the bipartite multigraph G(f) on vertex set M(f) ⊔ K(f) with an edge of multiplicity |m∩k| between each m∈M(f) and k∈K(f), add a ground vertex g joined by a single edge to every k∈K(f), let tau(f) = number of spanning trees of (G(f)∪{g}) computed via the matrix-tree theorem on the reduced Laplacian, and put sigma(f) = log_2(1+tau(f)). Then sigma(f) ≤ D_KW(f) · (|M(f)|+|K(f)|+1) · log_2(n+2), with the constant 1 in front being the conjectured tight prefactor. A single monotone f exhibiting sigma(f) strictly above this bound (using any verified upper bound on D_KW(f) from an explicit protocol) falsifies the conjecture.

Rationale

The sandpile group of the KW bipartite intersection multigraph counts independent 'cycles of certificates' that any KW protocol must resolve; each protocol step (a coordinate query at an internal node) can collapse at most O((|M|+|K|) log n) such independent cycles by a Cheeger-style averaging on the Laplacian, suggesting depth lower-bounds via log|K(G)|. Because tau(f) is determined by the BOTH-sided minterm/maxterm relation rather than a single-number truth-table statistic, the invariant is relational and exponential-time in the truth table, so it sidesteps the Razborov-Rudich natural-proofs barrier and the ring-only algebrization trap.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [2107.05128v2] Karchmer-Wigderson Games for Hazard-free Computation - [1012.2478v2] Packing spanning trees and the k-tree protocol - [0904.3087v4] Distributed Maintenance of Anytime Available Spanning Trees in Dynamic Networks - [1606.05050v1] Proof Complexity Lower Bounds from Algebraic Circuit Complexity - [1904.05483v2] Parallels Between Phase Transitions and Circuit Complexity?

Empirical Test

- exit code: `124`, elapsed: `240.10s`

```
TIMEOUT after 240s
```

Judge reasoning

The test timed out after 240s (returncode 124) without producing a RESULT line, so the pre-registered support condition cannot be evaluated and the critic challenge stands. | next: Optimize the test harness (e.g., cache D_KW upper bounds, reduce seed count or cap n at 6) so all 35 instances complete within the time budget.

Conway Temperature Bounds Tree-Resolution Width for 3-SAT

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Conway-Berlekamp combinatorial game theory: left/right stops and temperature of finite alternating scoring games (Conway 1976; Berlekamp-Wolfe 1994; Siegel 2013). This corner of CGT has essentially no presence in proof complexity outside Beck's positional-game tradition; a focused arXiv search for 'Conway temperature' AND ('resolution' OR 'CNF' OR 'proof complexity') returns no hits. × Tree-like Resolution refutation width w*(F) and size s*(F) for unsatisfiable 3-CNFs, viewed in the Cook-Reckhow-Krajicek bounded-arithmetic framework where ¬F is a Σ^b_0 statement whose $V^{0/S}$1_2 witnessing complexity is tied by Ben-Sasson–Wigderson to Resolution width and by Pudlak–Impagliazzo to the Prover–Adversary game.
- **Recorded**: 2026-05-17 06:17 UTC
- **Entry ID**: `18df83a46e71`

Statement

For unsat 3-CNF F on n ≤ 14 variables, define the alternating integer-scoring game G(F): players A (Maximizer) and B (Minimizer) alternate; on each turn the active player picks one unassigned variable and freely assigns a value; when all variables are assigned the score equals the number of falsified clauses. Compute the Conway left stop L(F) (Maximizer to move) and right stop R(F) (Minimizer to move) by memoized minimax on the full 2^n game tree, and let t(F) := (L(F)−R(F))/2 be the (degree-0) Conway temperature. Conjecture: (i) t(F) ≤ C_1 · w*(F) for every unsat 3-CNF F, where w*(F) is the tree-Resolution refutation width, and (ii) t(F) ≥ C_2 · n for Tseitin formulas T(G,ω) on 3-regular expander graphs G with v ≤ 10 vertices; a single instance violating either bound refutes the conjecture (defaults C_1 = 5, C_2 = 0.3).

Rationale

Tree-Resolution width is controlled by adversarial variable choice in the Prover-Adversary game, while the Conway temperature measures opposite-sign oscillation in optimal play of a scoring game — a structurally distinct invariant never (to our knowledge) computed on SAT-derived games. Tseitin formulas on expanders have Ω(n) resolution width by parity-flow, and the parity structure should also force balanced left/right moves at every vertex-clause group, yielding high temperature. If the bridge holds it gives a witnessing-style certificate of width inside the bounded-arithmetic Σ^b_0 Cook-Reckhow framework.

Novelty

- Judge: `NOVEL` over 6 arXiv hits

Top hits consulted: - [1709.03479v2] Conway's potential function via the Gassner representation - [2203.14669v1] Dynamic Structure in Four-strategy Game: Theory and Experiment - [2207.14140v1] Playing a 2D Game Indefinitely using NEAT and Reinforcement Learning - [2501.08313v1] MiniMax-01: Scaling Foundation Models with Lightning Attention - [2506.13585v1] MiniMax-M1: Scaling Test-Time Compute Efficiently with Lightning Attention

Empirical Test

- exit code: `0`, elapsed: `4.20s`


```
: 1, 'conjecture_holds': False, 'counterexample': 'Random 3-CNF with n=10 violated t(F) ≤ 5·w*(F): t=6
TRIAL: {'metric_name': 'temperature_to_width_ratio', 'metric_value': 0, 'instances_tested': 2, 'conj
CNF with n=10 violated t(F) ≤ 5·w*(F): t=6.5, w*=True'}
TRIAL: {'metric_name': 'temperature_to_width_ratio', 'metric_value': 0, 'instances_tested': 1, 'conj
CNF with n=10 violated t(F) ≤ 5·w*(F): t=6.5, w*=True'}
TRIAL: {'metric_name': 'temperature_to_width_ratio', 'metric_value': 0, 'instances_tested': 2, 'conj
CNF with n=10 violated t(F) ≤ 5·w*(F): t=6.5, w*=True'}
TRIAL: {'metric_name': 'temperature_to_width_ratio', 'metric_value': 0, 'instances_tested': 2, 'conj
CNF with n=14 violated t(F) ≤ 5·w*(F): t=9.5, w*=True'}
TRIAL: {'metric_name': 'temperature_to_width_ratio', 'metric_value': 0, 'instances_tested': 2, 'conj
CNF with n=10 violated t(F) ≤ 5·w*(F): t=6.0, w*=True'}
TRIAL: {'metric_name': 'temperature_to_width_ratio', 'metric_value': 0, 'instances_tested': 1, 'conj
CNF with n=10 violated t(F) ≤ 5·w*(F): t=6.0, w*=True'}
TRIAL: {'metric_name': 'temperature_to_width_ratio', 'metric_value': 0, 'instances_tested': 1, 'conj
CNF with n=8 violated t(F) ≤ 5·w*(F): t=5.5, w*=True'}
TRIAL: {'metric_name': 'temperature_to_width_ratio', 'metric_value': 0, 'instances_tested': 1, 'conj
CNF with n=8 violated t(F) ≤ 5·w*(F): t=6.0, w*=True'}
TRIAL: {'metric_name': 'temperature_to_width_ratio', 'metric_value': 0, 'instances_tested': 1, 'conj
CNF with n=8 violated t(F) ≤ 5·w*(F): t=5.5, w*=True'}
TRIAL: {'metric_name': 'temperatu
```


Judge reasoning

The 'w' *field is reported as the boolean True rather than an integer tree-Resolution width, so the comparison t(F) ≤ 5·w*(F) is being evaluated against a coerced value of 1 rather than the actual refutation width. The reported falsifications are artifacts of a broken width oracle, not genuine counterexamples. | next: Fix the tree-Resolution width computation to return an integer (exact tree-DPLL depth/width), re-run the 30-seed protocol, and additionally verify the Tseitin lower-bound clause se

---

Hochster Regularity of Variable Co-Occurrence Graph Lower-Bounds Resolution Width

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Castelnuovo–Mumford regularity of squarefree monomial edge ideals via Hochster's combinatorial formula on independence complexes (Hochster 1977; Bruns–Herzog 1993; Katzman 2006; Woodroofe 2014) — a corner of computational commutative algebra with essentially no proof-complexity-side publications (arXiv search 'edge ideal regularity' AND ('resolution width' OR 'proof complexity') returns no hits) × Resolution refutation width w*(F) for unsatisfiable 3-CNFs, viewed as a Ben-Sasson–Wigderson and Krajícek-style stepping stone toward size lower bounds for tree-like Resolution and Extended Frege (via the depth-O(1) Frege simulation of Resolution)

- **Recorded**: 2026-05-17 06:44 UTC
- **Entry ID**: `e0a06bc1af46`

Statement

For an unsatisfiable 3-CNF F on n ≤ 10 variables, build the variable co-occurrence graph G(F) with V = {x_1,…,x_n} and edges {x_i,x_j} whenever x_i and x_j co-appear in at least one clause of F. Define reg(F) := max{ j ≥ 0 : ∃ W ⊆ V with reduced rational homology H̃*{|W|−j−1}(Δ*{G(F)[W]}; Q) ≠ 0 }, where Δ_{G[W]} is the independence complex of the induced subgraph G[W] (Hochster's formula for reg(Q[x]/I_{G(F)})). Conjecture: for every unsatisfiable 3-CNF F, the minimum Resolution refutation width satisfies w*(F) ≥ reg(F) + 2; equivalently, a single F with w*(F) < reg(F) + 2 falsifies it.

Rationale

Hochster's formula recasts an algebraic invariant of the edge ideal as the maximum homological complexity over induced subgraphs, identifying variable clusters whose mutual logical entanglement is topologically nontrivial; intuitively, no narrow refutation can simultaneously break such a cluster, forcing high resolution width. The bridge from edge-ideal regularity to proof complexity is essentially unstudied, and unlike single-number truth-table invariants (calibration trap (j)), reg(F) is a *relational* invariant of the CNF graph computed by quantifying over induced substructures, sidestepping the Razborov–Rudich largeness criterion that defeats truth-table approaches. If valid, BSW gives size ≥ $\exp(\Omega(\mathrm{reg}(F)^2/n))$, and via Krajícek's depth-d Frege simulation this propagates to nontrivial EF line-count certificates.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.15s`

```
TRIAL: {'metric_name': 'w*(F) - reg(F)', 'metric_value': 0, 'instances_tested': 1, 'conjecture_holds'

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0d752578.py", line 149, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0d752578.py", line 133, in run_trial
    w = resolution_width(F, n)
        ^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0d752578.py", line 108, in resolution_w
   if any((var, True) in c1 and (var, False) in c2) or any((var, False) in c1 and (var, True) in c2):
          ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
TypeError: 'bool' object is not iterable
```

Judge reasoning

The test crashed with a TypeError in resolution_width (misuse of any() on a boolean expression), producing only 1 trial instead of the pre-registered 90 seeds. No valid data was collected to evaluate the support condition. | next: Fix the resolution_width function (the any() calls wrap non-iterable boolean expressions) and rerun the full 90-seed sweep across n∈{6,8,10}.

Ihara Zeta Entropy Predicts Tseitin DPLL Tree Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Ihara zeta function of finite graphs (Hashimoto–Stark–Terras combinatorial graph-zeta theory rooted in number-theoretic L-functions; essentially no papers connect Ihara zeta values to proof complexity, distinct from non-backtracking spectral approaches) × Tree-like Resolution / DPLL refutation tree size for Tseitin XOR formulas T(G,ω) on 3-regular graphs (Urquhart–Ben-Sasson–Razborov–Atserias regime)
- **Recorded**: 2026-05-17 07:23 UTC
- **Entry ID**: `3ba5471eff37`

Statement

Let G be a connected 3-regular graph on n vertices with adjacency matrix A, and let $\omega: V \to \{0,1\}$ have $\Sigma\omega$ odd, so T(G,ω) is unsatisfiable. Define the Ihara entropy $H(G) := (1/n)\cdot\ln\det((11/9)\cdot I - A/3)$, which is well-defined and positive because $(11/3)\cdot I - A$ has eigenvalues in [2/3, 20/3] for any 3-regular G. We conjecture that with pre-registered absolute constants $c_1 = 1/20$ and $c_2 = 20$, every connected 3-regular G with odd ω satisfies $c_1\cdot n\cdot H(G) - 1 \le \log_2(S\_DPLL(T(G,\omega))) \le c_2\cdot n\cdot H(G) + \log_2(n) + 1$, where S_DPLL is the leaf count of the canonical DPLL refutation tree using unit propagation plus first-unassigned-variable branching. A single (G,ω) violating either inequality refutes the conjecture.

Rationale

The Ihara zeta function $\zeta\_G(u) = \prod\_p (1-u^{\{|p|\}})\{-1\}$ over prime non-backtracking cycles is encoded by $\det(I - uA + u^2(D-I))$; at u = 1/3 this determinant aggregates how cycles of all lengths contribute multiplicatively, and each cycle of G corresponds in T(G,ω) to a closed XOR constraint that DPLL must locally certify, so per-vertex cycle entropy should track per-vertex branching entropy. The evaluation point u = 1/3 sits strictly inside the radius of convergence for 3-regular graphs, giving a numerically stable, integer-matrix invariant whose growth on Ramanujan-type expanders matches Urquhart's exponential refutation-size lower bound. This bridges number-theoretic graph zetas to proof complexity in a regime where, to our knowledge, fewer than five papers have looked.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `240.01s`

```
TIMEOUT after 240s
```

Judge reasoning

The test timed out after 240s (returncode=124) without producing a RESULT line or any instance data, so neither the bound inequalities nor the Spearman ρ threshold could be evaluated. | next: Reduce the instance budget (e.g. fewer seeds or cap n at 12) and/or replace the exact DPLL leaf count with an iterative bounded search so the 150-instance sweep completes within the timeout.

Signed-Support L_inf Discrepancy Bounds Bottom Fan-In for AC0 PARITY

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Combinatorial discrepancy theory (Beck-Fiala, Spencer six-standard-deviations, Chazelle-Matousek hereditary discrepancy of ±1 incidence matrices) × AC^0 depth-d circuit size and bottom fan-in for PARITY_n, in the Hastad / Yao switching-lemma regime

- **Recorded**: 2026-05-17 08:06 UTC
- **Entry ID**: `2f193a8b2209`

Statement

For an AC^0 circuit C on n inputs with bottom-level AND/OR gates B_1,...,B_m, define the signed support matrix M(C) in {-1,0,+1}^{m x n} by M_ij = +1 if x_j appears positively in B_i, -1 if negatively, and 0 if absent, and let w(C) = max_i ||M_i||*0 be the maximum bottom fan-in. Define the L_infty discrepancy psi(C) := min*{chi in {-1,+1}^n} max_{i<=m} | sum_j M_ij chi_j |. CONJECTURE: there is an absolute constant c >= 1/4 such that for every AC^0 circuit C of any constant depth that computes PARITY_n, psi(C) >= c * w(C); a single PARITY-computing AC^0 instance with psi(C) < (1/4) * w(C) refutes it.

Rationale

PARITY forces every bottom-gate sign pattern to participate in a covering of (n-1)-parity-correct assignments, so the rows of M(C) must span a Hadamard-like sign system that admits no balanced coloring whose row-sums all fall below the typical sqrt(w) scale; this is a structural cancellation obstruction distinct from random restrictions. The invariant is exactly S_n x Z_2^n equivariant (variable relabeling and literal-polarity flips both preserve psi and w), satisfying the focus-mode naturality demand. If correct, the bound psi >= c*w replaces the switching-lemma probability calculation by a deterministic discrepancy certificate and is sharp on the canonical CNF/DNF (where psi ~ n = w).

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `240.10s`

```
TIMEOUT after 240s
```

Judge reasoning

Test timed out after 240s (returncode 124) before producing any RESULT line or data; the pre-registered support condition cannot be evaluated. | next: Reduce search space (e.g., restrict to n in {4,6,8} and fewer seeds) or optimize psi computation via branch-and-bound/SDP relaxation to obtain data within the time budget.

2-Sylow Rank of Critical Group Bounds Tseitin DPLL Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Algebraic combinatorics of the critical (sandpile) group K(G): specifically the 2-Sylow rank r_2(G)=dim_{F_2} ker(L̃_G mod 2) of the reduced graph Laplacian, in the Bjorner–Lovasz–Shor / Wood–Mészáros tradition on cokernels of random integer matrices; distinct from the well-studied invariant |K(G)|=τ(G) and essentially absent from Tseitin proof-complexity literature × Tree-like Resolution / DPLL refutation size for Tseitin XOR formulas T(G,ω) on 3-regular graphs (Urquhart–Ben-Sasson–Razborov regime), via the cycle-space-mod-2 mechanism underlying expander Tseitin lower bounds
- **Recorded**: 2026-05-17 08:50 UTC
- **Entry ID**: `eb9e15c9c9db`

Statement

For every connected 3-regular graph G on n vertices and every charge ω:V→F_2 with Σω≡1 (mod 2), let L̃*G be the reduced graph Laplacian (Laplacian with one row/column removed) and define the 2-Sylow rank r_2(G):=(n−1)−rank*{F_2}(L̃_G mod 2). Let N_DPLL(T(G,ω)) be the number of nodes in the canonical DPLL refutation tree of the Tseitin CNF (with unit propagation, branch on lowest-index unassigned variable). Conjecture: log_2 N_DPLL(T(G,ω)) ≥ (1/4)·(n−1−r_2(G)) for every such (G,ω); equivalently, a single (G,ω) with log_2 N_DPLL < (1/4)(n−1−r_2(G)) falsifies it.

Rationale

Tseitin refutation hardness is governed by the cycle-space mod 2 of G, and r_2(G) precisely captures how many independent F_2-degenerate directions the Laplacian pairing has on that cycle space; thus low r_2(G) (full F_2-rigidity) should force DPLL to traverse all cycle-space cosets while high r_2 opens algebraic short-cuts. This invariant is graph-structural (not a truth-table predicate), bypassing natural proofs, and is computed by elementary F_2 linear algebra rather than ring extensions, so it does not algebrize. It connects an under-utilised corner of algebraic combinatorics (2-part of Jacobian cokernel distributions) to the established expander-Tseitin lower-bound machinery.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `240.10s`

```
TIMEOUT after 240s
```

Judge reasoning

The test timed out after 240s (returncode 124) without producing a RESULT line or any instance data, so the pre-registered support condition over 150 instances cannot be evaluated. | next: Re-run with a longer timeout and/or reduce the size range (e.g., n∈{6,8,10,12}) and cache DPLL counts to obtain complete data on all 150 instances.

Stable-Rank of Layer-Symbol Matrix Separates Read-Twice BPs for IP_2

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Forster-style stable-rank / numerical-rank theory (Forster 2002; Linial–Shraibman 2009) — a corner of sign-rank / discrepancy-based lower-bound machinery transplanted from communication matrices to the operator-valued symbol bundle of a branching program; this bridge has essentially no presence in BP literature outside the unrelated Nechiporuk rank tradition. × Read-twice oblivious branching program size for IP_2 on 2n adversarial-ordered variables (Borodin–Razborov–Smolensky 1993 open separation), in the meta-complexity sense of seeking a 'compressibility witness' ρ(P) with ρ(P)=O(log size(P)) — a poly-time-computable structural invariant of the BP, NOT of its truth-table.
- **Recorded**: 2026-05-17 09:02 UTC
- **Entry ID**: `46c3da6284c4`

Statement

For a read-twice oblivious BP P of size s on 2n variables, with layers t=1..4n and transition matrices M_t(b)∈{0,1}^{s×s} for b∈{0,1}, form the layer-symbol matrix S(P) := [Δ_1 | Δ_2 | … | Δ_{4n}]

$\in \mathbb{R}^{s \times 4n \cdot s}$ where $\Delta\_t := M\_t(1) - M\_t(0)$. Define the stable-rank invariant $\rho(P) := \|S(P)\|\_F^2 / \|S(P)\|\_op^2$. Conjecture: (i) $\rho(P) \le 2 \cdot s$ for every read-twice BP P (numerical-rank anchor: this follows from rank $\le$ s but we conjecture the tighter constant 2 holds for all syntactic BPs); (ii) for every read-twice BP P with output correlation $\langle f\_P, IP\_2\_n \rangle \ge \varepsilon$ over uniform 2n-bit inputs, $\rho(P) \ge c \cdot \varepsilon^2 \cdot 2^{n/2}$ for an absolute constant $c > 0$. A single instance with $\langle f\_P, IP\_2 \rangle \ge \varepsilon$ and $\rho(P) < c \cdot \varepsilon^2 \cdot 2^{n/2}$, or any read-twice BP with $\rho(P) > 2s$, refutes it.

Rationale

Each Δ_t records how flipping one read at layer t shifts the BP's state operator; for a small BP these shifts share an s-dimensional column space, capping the stable rank at O(s). IP_2's adversarial pairing of x_i with y_i forces the symbol bundle to encode $2^{\Theta(n)}$ mutually-orthogonal cross-variable shift directions (after the middle cut, each pair (x_i,y_i) contributes a Hadamard-flavoured outer product), so the Frobenius mass spreads across exponentially many comparable singular values, making $\|S\|\_F^2/\|S\|\_op^2$ grow geometrically with n. The setup avoids natural-proofs (ρ is a structural property of P, not its truth-table), algebrization (singular values are analytic, not ring-arithmetic), and relativization (no oracle).

Novelty

- Judge: `NOVEL` over 6 arXiv hits

Top hits consulted: - [0710.5926v2] Mod 2 cohomology of 2-local finite groups of low rank - [1503.07648v2] Sign rank versus VC dimension - [1708.02054v1] Pseudorandom Bits for Oblivious Branching Programs - [0809.2093v2] An approximation algorithm for approximation rank - [1706.02207v3] On The Communication Complexity of High-Dimensional Permutations

Empirical Test

- exit code: `0`, elapsed: `0.08s`


```
551269602595 > 2s=8", "seed": "421"}
TRIAL: {"metric_name": "stable_rank", "metric_value": 87.00000000000001, "metric_std": 0.0, "instanc
TRIAL: {"metric_name": "stable_rank", "metric_value": 80.24174837076876, "metric_std": 0.0, "instanc
TRIAL: {"metric_name": "stable_rank", "metric_value": 39.25023292246046, "metric_std": 0.0, "instanc
TRIAL: {"metric_name": "stable_rank", "metric_value": 47.12182530062137, "metric_std": 0.0, "instanc
TRIAL: {"metric_name": "stable_rank", "metric_value": 65.81197395850668, "metric_std": 0.0, "instanc
TRIAL: {"metric_name": "stable_rank", "metric_value": 0.0, "metric_std": 0.0, "instances_tested": 0,
TRIAL: {"metric_name": "stable_rank", "metric_value": 36.561209216717984, "metric_std": 0.0, "instan
TRIAL: {"metric_name": "stable_rank", "metric_value": 28.50532667271179, "metric_std": 0.0, "instanc
TRIAL: {"metric_name": "stable_rank", "metric_value": 59.148813421486146, "metric_std": 0.0, "instan
```


Judge reasoning

The critic correctly notes that $\rho(P)=\|S\|\_F^2/\|S\|\_op^2$ is mathematically bounded by rank(S) ≤ s, so reported values like 26–129 with 2s=8 are impossible and indicate a metric-implementation bug (likely summing per-layer ratios instead of computing the stable rank of the concatenated matrix). The pre-registered support condition is therefore not validly evaluated, and the 'falsification' cannot be trusted. | next: Fix the ρ computation to actually compute $\|S\|\_F^2/\|S\|\_op^2$ on the concatenated S∈ℝ^{s×4n

---

TP_2 Minor-Sign Defect of Spectral Pseudo-Moment Matrix Bounds Max-CUT Ratio

- **Verdict**: `INCONCLUSIVE`

- **Bridge**: Karlin–Edrei–Schoenberg total positivity / Pólya frequency theory (TP_k matrices, sign-regularity of 2x2 minors) × Degree-2 SOS pseudo-moment matrix for Max-CUT on graphs (Goemans–Williamson regime), with a lifting interpretation via gadget composition
- **Recorded**: 2026-05-17 09:29 UTC
- **Entry ID**: `16809e692923`

Statement

For a connected graph G=(V,E) on n vertices with combinatorial Laplacian L=D-A, let v_2,v_3∈R^n be the second and third eigenvectors of L (unit-norm, breaking ties by lexicographic sign). Define the rank-2 spectral pseudo-moment matrix M_G[i,j]:=(v_2[i]v_2[j]+v_3[i]v_3[j])/Z, where Z:=max_{a,b}|v_2[a]v_2[b]+v_3[a]v_3[b]|, and the TP_2 defect δ(G):=|{(i<j,k<l):det M_G[{i,k},{j,l}]<-1/n$^{3\}|/\mathrm{binom}(n,2)}$2. Conjecture: there exist absolute constants c_1=0.5 and c_2=0.10 such that (i) every graph with max-CUT(G)/|E|≥0.879 satisfies δ(G)≤c_1·n^{-1/2}, and (ii) every graph on n≥12 vertices with max-CUT(G)/|E|≤0.55 satisfies δ(G)≥c_2. A single graph violating either inequality refutes the conjecture.

Rationale

Pólya frequency / TP_2 matrices are exactly the moment matrices of log-concave one-parameter families; SOS-tight max-CUT instances (where degree-2 SOS already certifies a near-optimal cut) should correspond to log-concave 1D rounding directions and thus preserve TP_2 in the leading spectral block, whereas integrality-gap instances spread mass in a way that flips 2x2 minor signs. Because TP_2 is closed under coordinatewise polynomial gadget tensoring (Karlin 1968, Ch. 3), δ behaves as a lifting-friendly invariant: gadget composition can only grow δ by a constant factor, matching the 'preserved under polynomial reductions' requirement on the target property P. The bridge between Pólya frequency theory and SOS Max-CUT lower bounds appears to have under five papers (TP_k has been used in stable-polynomial proofs of Kadison–Singer but not in SOS degree-lower-bound machinery).

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `3.25s`

```
TRIAL: {'metric_name': 'TP_2 defect', 'metric_value': 0.0004911287328869746, 'instances_tested': 8,

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_41095b21.py", line 253, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_41095b21.py", line 216, in run_trial
    delta = tp2_defect(graph)
            ^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_41095b21.py", line 186, in tp2_defect
    M = spectral_pseudo_moment_matrix(graph)
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_41095b21.py", line 174, in spectral_pse
    eigenvalues, eigenvectors = matrix_eigen(L)
                                ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_41095b21.py", line 80, in matrix_eigen
    Q, R = qr_decomposition(A)
           ^^^^^^^^^^^^^^^^^^^
```

```
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_41095b21.py", line 96, in qr_decomposit
    Q[i][k] = A[i][k] / R[k][k]
              ~~~~~~~~^~~~~~~~~
ZeroDivisionError: float division by zero
```

Judge reasoning

The test crashed with a ZeroDivisionError in the custom QR decomposition after only 8 instances, well short of the pre-registered ~600 instance protocol. Although a putative low-cut counterexample was reported, the run terminated abnormally and the support condition cannot be evaluated. | next: Replace the hand-rolled QR with numpy.linalg.eigh (or guard against zero pivots) and rerun the full 4×30×5 protocol before judging the conjecture.

Gromov 4-Point δ of XOR-Lifted Row Metric Bounds DT Depth

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Coarse geometry / Gromov 4-point δ-hyperbolicity of finite metric spaces (Gromov 1987; Bridson–Haefliger 1999; Coornaert 2014) — a relational metric invariant from geometric group theory with essentially no presence in communication complexity (search 'Gromov hyperbolicity' AND 'communication complexity' returns no direct hits; <5 papers in adjacent areas) × Deterministic two-party communication complexity of XOR-gadget-lifted Boolean functions, in the Raz–McKenzie / Göös–Pitassi–Watson / Hatami–Hossein-Pitassi query-to-communication lifting framework, accessed through the bipartite row metric of the lifted communication matrix M_{f∘XOR_2}
- **Recorded**: 2026-05-17 10:01 UTC
- **Entry ID**: `5ee62d63ba3c`

Statement

Let f:{0,1}^n→{0,1} have deterministic decision-tree depth d=DT(f), let g=XOR_2 (Alice has x_i∈{0,1}, Bob has y_i∈{0,1}, g(x_i,y_i)=x_i⊕y_i), and let M∈{0,1}$^{\{2}$n×2^n} be the lifted matrix M[x,y]=f(x⊕y). View rows of M as points in ({0,1}$^{\{2}$n},d_H). For each 4-tuple of distinct rows (a,b,c,d) compute the three sums S_1=d_H(a,b)+d_H(c,d), S_2=d_H(a,c)+d_H(b,d), S_3=d_H(a,d)+d_H(b,c), sort them S_{(1)}≥S_{(2)}≥S_{(3)}, and set δ(M)=max over 4-tuples of (S_{(1)}−S_{(2)})/2; let δ(M)=δ(M)/diam(M). Conjecture: for every non-constant f, δ(M_{f∘XOR_2}) ≤ 1 − DT(f)/(n+1), i.e. higher DT-depth forces strictly more 4-point-treelike row geometry; a single f with δ(M) > 1 − DT(f)/(n+1) refutes it.

Rationale

Lifting theorems claim DT structure of f survives into the gadget-amplified bipartite matrix; if so, the inherited combinatorial structure should constrain coarse-geometric quantities of the row Hamming cloud, since high-DT functions composed with XOR_2 build hierarchical query trees that the Gromov 4-point inequality is precisely designed to detect. Crucially δ is a 4-point relational invariant — not a single-number truth-table functional — so it sidesteps the Razborov–Rudich largeness obstruction, and uses only Hamming distances (no ring/polynomial structure), so it should not algebrize. If the bound holds, it gives a coarse-geometric proxy for the lifted communication lower bound via embedding-distortion arguments à la Linial–London–Rabinovich.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f5a45145.py", line 97, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f5a45145.py", line 69, in run_trial
    s3 = distances[sum(range(a+1, d+1))] + distances[sum(range(b+1, c+1))]
         ~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^
IndexError: list index out of range
```

Judge reasoning

The test crashed with an IndexError before producing any data, so neither the support condition nor a counterexample was established. Per the rules, a crash yields INCONCLUSIVE. | next: Fix the indexing bug in the 4-tuple distance lookup (use a proper pair-to-index mapping like distances[(i,j)] or itertools.combinations indexing) and rerun the pre-registered sweep.

Median Half-Subgraph Separator Lower-Bounds Tseitin DPLL Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Separation profile of geometric group theory (Benjamini–Schramm–Timár 2012; Hume–Mackay–Tessera 2020; Hume 2017): the coarse-geometric invariant sep_G(k) = sup over k-vertex subgraphs H of min balanced vertex separator of H, used to distinguish quasi-isometry classes of finitely generated groups. This corner of GGT has essentially no presence in proof complexity — an arXiv search 'separation profile' AND ('resolution' OR 'Tseitin' OR 'proof complexity') returns no hits; the closest neighbours are asymptotic-dimension papers that never reach proof systems. × Tree-like Resolution / DPLL refutation tree size for Tseitin XOR formulas T(G,ω) on 3-regular graphs, in the Urquhart–Ben-Sasson–Wigderson–Krajíček bounded-arithmetic regime where ¬T(G,ω) is a Σ^b_0 statement and DPLL size lower-bounds V^0-witnessing complexity.
- **Recorded**: 2026-05-17 10:32 UTC
- **Entry ID**: `118fd4951a23`

Statement

For a connected 3-regular graph G on m vertices and odd charge ω: $\nu\_30(G)$:= median over 30 uniformly random vertex subsets U ⊂ V(G) of size k=⌊m/2⌋ of the value β(G[U]), where β(H) is the size of a minimum vertex separator (A,V(H)∖(A∪S),S) of G[U] with |A|,|V(H)∖(A∪S)| ≥ ⌊|V(H)|/3⌋, and β(H):=0 if G[U] is disconnected. Conjecture: (i) every odd-charge Tseitin formula T(G,ω) requires DPLL tree size s*(T(G,ω)) ≥ 2^{0.1·ν_30(G)}; (ii) for every 3-regular G admitting a vertex partition (V_1,V_2) with |V_1|,|V_2| ≥ m/3 and |E(V_1,V_2)| ≤ 2 (e.g., barbells, prism-pairs joined by a bridge or two edges), ν_30(G) ≤ 2. A single 3-regular instance with log_2(s*) < 0.1·ν_30(G), or a small-cut graph in class (ii) with ν_30(G) > 2, refutes the conjecture.

Rationale

The Ben-Sasson–Wigderson width method bounds Tseitin Resolution length via expansion of G *as a whole*; but bounded arithmetic V^0-witnessing of unsatisfiable Σ^b_0-instances should react to *every scale* of expansion, not just the global one. The separation profile evaluated at half-size captures exactly this: it is large iff *typical* half-subgraphs of G are themselves robustly connected, which is the structural data DPLL must repeatedly re-encounter on any branch that fixes half the variables. The link to Cook-Reckhow-Krajíček: ν_30 is a low-complexity (NC-computable up to sampling) Σ^b_0-definable property of G, so a separation 'ν_30 large ⇒ refutation long' is exactly the bounded-arithmetic style of structural lower bound that escapes both NATURAL_PROOFS (ν_30 depends on G, not on a truth-table) and ALGEBRIZATION (no ring extension respects vertex-separator counts).

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_41e6d60a.py", line 194, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_41e6d60a.py", line 155, in run_trial
    nu_30 = compute_nu_30(edges, n)
            ^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_41e6d60a.py", line 68, in compute_nu_30
    nu_values.append(compute_min_vertex_separator(subgraph_edges, k))
                     ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_41e6d60a.py", line 46, in compute_min_v
    if not is_connected(edges, n):
           ^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_41e6d60a.py", line 35, in is_connected
    graph[v].add(u)
    ~~~~~^^^
KeyError: 7
```

Judge reasoning

The test crashed with a KeyError in is_connected before producing any RESULT line, so neither support nor falsification conditions can be evaluated. The critic also challenged due to lack of data. | next: Fix the graph construction in is_connected (initialize adjacency entries for all vertices in the induced subgraph, e.g., using defaultdict(set) or pre-populating keys) and rerun the pre-registered protocol.

Cubic-Form Lie-Stabilizer Codim Lower-Bounds DISJ Communication

- **Verdict**: `BARRIER_HIT`
- **Bridge**: Geometric Complexity Theory: Mulmuley–Sohoni Lie-algebra stabilizers of explicit cubic forms c_M ∈ Sym$^{3(R}$N), accessed through the codimension $\kappa(M)=N^2 - \dim$ Stab_{gl_N}(c_M). This is a first-order/infinitesimal GCT invariant distinct from (a) Kronecker-coefficient counts and (b) full orbit-closure dimensions, the two GCT-COMM bridges that previously hit ALGEBRIZATION in this approach; <5 papers connect cubic-form stabilizers to communication complexity. × Randomized two-party communication complexity CC_R(M) of Boolean matrices M ∈ {0,1}^{N×N}, with the target M = M_DISJ_n on N = 2^n inputs (Razborov 1992 Ω(n) bound), instantiating the COMM_DISJ tensor-invariant program.
- **Recorded**: 2026-05-17 10:38 UTC
- **Entry ID**: `9d129ecb4a7f`

Statement

For $M \in \{0,1\}^{N\times N}$ ($N = 2^n$), form the symmetric cubic form $c_M(x) := \Sigma_{i,j,k\in[N]} M[i,j]\cdot M[j,k]\cdot M[i,k]\cdot x_i x_j x_k$ whose coefficient at the symmetrized monomial $x_i x_j x_k$ is the count of directed triangles (i,j,k) in the bipartite-merged graph of M. Let $g(M) := \dim\{A \in gl_N$: the polynomial $\Sigma_i (Ax)_i \cdot \partial c_M/\partial x_i$ is identically zero$\}$, and define the GCT stabilizer codimension $\kappa(M) := N^2 - g(M)$. Conjecture: (a) for every Boolean M, $CC_R(M) \geq (1/8)\cdot\log_2(1 + \kappa(M))$, refuted by any single matrix with a randomized log-rank-style upper bound $CC_R(M) \leq 2\cdot\log_2(\text{rank}_R(M)) + 1$ violating $(1/8)\cdot\log_2(1+\kappa(M))$ by more than 1; (b) $\kappa(M_{DISJ_n}) \geq N - N^{3/4}$ for all $n \geq 3$, so (a) certifies $CC_R(DISJ_n) \geq n/8 - O(1) = \Omega(n)$.

Rationale

The triangle-count cubic form c_M encodes a non-polynomial-extension-friendly combinatorial feature of M (counting 3-cycles in its associated digraph) that is invisible to algebraic oracles, addressing why prior Kronecker/orbit-dimension GCT-COMM attempts hit ALGEBRIZATION. The infinitesimal Lie stabilizer is a first-order Mulmuley–Sohoni invariant that, unlike rank, is sensitive to triangle-richness — and DISJ's communication matrix has a sparse 3-disjoint structure ($|\{(S,T,U)$: pairwise disjoint$\}| = 3^n$ □ N^3) that should generically force a trivial stabilizer. The 1/8 constant tracks a sym-cubic Hilbert-series degeneracy factor and is conservative.

Novelty

- Judge: " over 0 arXiv hits

Empirical Test

- exit code: `-1`, elapsed: `0.00s`

```
(no output)
```

Judge reasoning

Rejected by ALGEBRIZATION barrier

Discrete Morse Critical Cells of Conflict Complex Bound DPLL Tree Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Discrete Morse theory on simplicial complexes (Forman 1998; Joswig–Pfetsch 2005 greedy lex matching; Benedetti–Lutz 2014 random discrete Morse) — a corner of combinatorial topology essentially absent from proof complexity (arXiv 'discrete Morse' AND ('Resolution' OR 'DPLL' OR 'proof complexity') returns no direct hits, with <5 adjacent independence-complex papers) × DPLL / tree-like Resolution refutation tree size $t(F)$ *for unsatisfiable 3-CNFs, in the Cook–Reckhow–Krajíček bounded-arithmetic regime where $\neg F$ is a Σ^b_0 statement and $\log_2$* $t(F)$ controls V^0-witnessing complexity of unsatisfiability
- **Recorded**: 2026-05-17 11:47 UTC
- **Entry ID**: `02fea1709cc2`

Statement

For an unsat 3-CNF F with clauses $c_1,\ldots,c_m$, let C_F be the clause-conflict graph (vertex per clause; edge $\{c_i,c_j\}$ iff they share at least one complementary literal pair) and let $I_{\leq 3}(C_F)$ be the independence complex of C_F truncated to faces of size ≤4 (faces = clause-subsets pairwise free of complementary-literal conflicts). Run Joswig–Pfetsch greedy lex discrete-Morse matching: traverse simplices in (dim, lex) order, pair each σ with its lex-smallest currently-unmatched coface τ of dim(σ)+1 inside the truncation (otherwise mark σ critical). Let δ(F) := number of critical cells of dimension ≥1. Conjecture: there is an absolute constant c ≥ 1/4 such that for every unsat 3-CNF F with m ≥ 4, log_2(t*(F)) ≥ c · log_2(δ(F)+2), where t*(F) is the DPLL refutation tree-leaf count under lex-variable branching with unit propagation. A single F with measured log_2(t*(F)) < 0.25·log_2(δ(F)+2) refutes it.

Rationale

Critical cells of I(C_F) count obstructions to extending a locally-consistent partial certificate (a set of mutually non-resolving clauses) to a global one, which is precisely the local-to-global obstruction Ben-Sasson–Wigderson width-size trade-offs quantify for DPLL. A logarithmic relation is the natural scale: t* grows exponentially in hardness while δ grows exponentially in conflict-topological richness, so log–log alignment is plausible. The bridge avoids algebrization (Forman matching is purely combinatorial — no ring, no polynomial extension), evades natural proofs (the invariant is syntactic on F, not on its truth table), and lies in a regime Razborov/Krajíček bounded-arithmetic witnessing leaves open.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `0`, elapsed: `0.04s`

```
7}
TRIAL: {'metric_name': 'log2(t*)/log2(δ+2)', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture
TRIAL: {'metric_name': 'log2(t*)/log2(δ+2)', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture
TRIAL: {'metric_name': 'log2(t*)/log2(δ+2)', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture
TRIAL: {'metric_name': 'log2(t*)/log2(δ+2)', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture
TRIAL: {'metric_name': 'log2(t*)/log2(δ+2)', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture
TRIAL: {'metric_name': 'log2(t*)/log2(δ+2)', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture
TRIAL: {'metric_name': 'log2(t*)/log2(δ+2)', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture
TRIAL: {'metric_name': 'log2(t*)/log2(δ+2)', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture
TRIAL: {'metric_name': 'log2(t*)/log2(δ+2)', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture
TRIAL: {'metric_name': 'log2(t*)/log2(δ+2)', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture
TRIAL: {'metric_name': 'log2(t*)/log2(δ+2)', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture
TRIAL: {'metric_name': 'log2(t*)/log2(δ+2)', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture
TRIAL: {'metric_name': 'log2(t*)/log2(δ+2)', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture
RESULT: INCONCLUSIVE reason=no_valid_instances
```

Judge reasoning

parse_fail | next: NONE

Gate-Cone Poset Mobius Value Lower-Bounds ACC0 Size for MOD-q

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Mobius functions on incidence algebras of finite posets (Rota 1964 combinatorial inversion; Stanley EC1 Ch.3) applied to the gate-dependency poset of an ACC^0 circuit; an algebraic-combinatorial invariant essentially absent from circuit complexity (arXiv search 'Mobius function poset' AND 'ACC^0' returns no hits; <5 adjacent papers on poset invariants for circuits) × ACC^0[m] circuit size for explicit symmetric functions (notably MOD_q for q coprime to m) in the Williams 2011 / Murray-Williams 2018 algorithmic-method regime, where structural circuit invariants underlie speedup-style lower bounds (NEXP ⊄ ACC^0)

- **Recorded**: 2026-05-17 15:23 UTC
- **Entry ID**: `3b280d479739`

Statement

Let C be an ACC^0[m] circuit on n inputs with s gates and depth d. Adjoin a formal minimum 0̂ below all gates and let P(C) be the poset on {0̂} ∪ Gates(C) with g ≤ h iff h = g or there is a directed wire path g → h (and 0̂ < g for every gate g). Define M(C) := |μ_{P(C)}(0̂, g_out)| where μ is the Mobius function of P(C) and g_out is the output gate, computed by μ(0̂,0̂)=1 and μ(0̂,h) = −Σ_{0̂ ≤ z < h} μ(0̂,z). Conjecture (A, upper anchor): M(C) ≤ (s+1)^d for every ACC^0[m] circuit. Conjecture (B, hardness anchor): for every ACC^0[m] circuit C that correctly computes MOD_q on n inputs with gcd(q,m)=1, one has M(C) ≥ $2^{\{c \cdot n}$\{1/(d+1)\}\} for an absolute constant c ≥ 1/8; a single counterexample circuit refutes (B).

Rationale

Mobius inversion on the gate-dependency poset is a CIRCUIT invariant (not a truth-table property), so it dodges the natural-proofs trap that killed the Plunnecke variant in this approach; it is also non-ring (no polynomial extension makes the recursion μ(0̂,h)=−Σμ(0̂,z) meaningful over F_p[X]), so it should not algebrize. Heuristically, Smolensky’s low-degree-polynomial obstruction for MOD_q in ACC^0[m] forces long, non-factorisable dependency chains in any small circuit, which inflate alternating-sign Mobius cancellations on antichains of the gate poset; the M(C) invariant captures this ‘inversion depth’ more sharply than gate count alone and is computable in O(s^2) time, making it a viable Williams-style structural witness.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.10s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_fbcfc841.py", line 123, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_fbcfc841.py", line 78, in run_trial
    circuit = generate_random_circuit(n, s, d, seed)
              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_fbcfc841.py", line 27, in generate_rand
    num_inputs = random.randint(2, min(3, len(gates)))
                 ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/usr/lib/python3.12/random.py", line 336, in randint
    return self.randrange(a, b+1)
           ^^^^^^^^^^^^^^^^^^^^^^
  File "/usr/lib/python3.12/random.py", line 319, in randrange
    raise ValueError(f"empty range in randrange({start}, {stop})")
ValueError: empty range in randrange(2, 1)
```

Judge reasoning

The test crashed with a ValueError in generate_random_circuit (empty range in randrange(2,1)) before any instances were evaluated, so neither the support nor falsification condition could be assessed. | next: Fix the circuit generator to guarantee len(gates) ≥ 2 (or clamp the fan-in lower bound) before sampling num_inputs, then rerun the pre-registered sweep.